



Atul Jha
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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology, Guwahati	7.26 (Current)	2022-Present
Senior Secondary	CBSE Board	88%	2022
Secondary	CBSE Board	93.6%	2020

PROJECTS

- Automated Multimodal Literature Review** Jan. 2024 - Feb. 2024
IITG.Ai club project GitHub
 - Integrated **text + image data**, improving query comprehension and response accuracy by **25%** compared to text-only baselines.
 - Implemented **Retriever-Augmented Generation (RAG)**, boosting answer **precision by 18%** and **coherence by 22%**.
 - Applied system across diverse domains including **education, research, and customer service** for versatile real-world impact.
 - Streamlined deployment with **Python, PyTorch, and Docker**, ensuring smooth installation and rapid project setup.
- Unconditional Image Generation Using Diffusion Model** May. 2025 - Jun. 2025
Personal project GitHub
 - Implemented **UNet** and **Attention-UNet** from scratch, achieving **15% lower FID score** with attention mechanisms.
 - Developed models in **PyTorch** and built an interactive **Streamlit** app for real-time image generation and model comparison.
 - Automated end-to-end data loading with **HuggingFace datasets** library, reducing manual preprocessing effort by nearly **60%**.
 - Enabled users to generate and visualize **batch outputs (up to 32 images)** for model comparison between variants in real-time.
- AI Image Classifier** Mar. 2025 - Apr. 2025
Personal project GitHub
 - Built a **web-app** to classify images as fake/real with **91% accuracy** by utilizing **P-score confidence** to detect AI edits.
 - Fine-tuned **InceptionV3** architecture, outperforming **VGG16** baseline by a **7% accuracy gain** on validation sets.
 - Applied **SMOTE** oversampling, improving minority class recall by **20%** on imbalanced datasets scraped with **Selenium**.
 - Deployed an interactive **Streamlit UI**, enabling users to upload images and receive **real-time predictions** in under **2 seconds**.
- News Sentiment Analysis Portal** May 2024 - Jun. 2024
Coding Club Even Sem project GitHub
 - Scraped **10,000+ real-time news headlines** using **Selenium**, allowing users to seamlessly filter and explore by **5+ domains**.
 - Compared **Logistic Regression, Random Forest, LSTM, and fine-tuned BERT**, achieving up to **87% accuracy** with BERT.
 - Performed **data preprocessing and feature engineering** to improve model accuracy and handle noisy news headlines.
 - Developed an interactive **Streamlit** dashboard enabling real-time browsing and analysis of **50+ headlines per refresh**.

TECHNICAL SKILLS

- Data Science/AI:** Pandas, NumPy, TensorFlow, PyTorch, Keras, scikit-learn, RAG, Fine Tuning
- ML Techniques:** Linear/Logistic Regression, SVM, KMeans, KNN, PCA, XGBoost, CatBoost, LightGBM
- Deep Learning:** Neural Networks, CNNs, GANs, RNNs, LSTMs, Transformers
- Data Visualization:** Matplotlib, Seaborn, Plotly, Tableau
- Programming Languages:** Python, C/C++, SQL
- Tools:** Git, Streamlit

KEY COURSES TAKEN

- Machine Learning.AI, IIT Guwahati:** Linear Regression, Logistic Regression, Regularization, Naïve Bayes, Evaluation Metrics, SVM, KMeans, KNN, unsupervised clustering, PCA, XGBoost, CatBoost and LightGBM.
- Deep Learning Specialization by DeepLearning.AI:** Neural Networks and Deep Learning, Improving Deep Neural Networks, Structuring Machine Learning Projects, Convolutional Neural Networks, Sequence Models.

ACHIEVEMENTS

- Amazon ML Summer School** – Selected for Amazon’s **flagship ML program** among **80,000** candidates. 2024
- Adobe Gensolve Hackathon** – Ranked in the top **5 percentile of participants**. 2024
- Kriti** – Secured **Silver** medal in **ML** problem statement in **Inter Hostel Tech** Competition. 2024
- WorldQuant** – Reached **Gold Level** by developing various trading strategies (**Alphas**). 2023
- JEE (Mains)** – Secured **All India Rank 8695 (99.04 percentile)** among 1 million students. 2022
- JEE (Advance)** – Secured **All India Rank 5768** among 0.2 million students. 2022

EXTRACURRICULARS

- Inter IIT:** Represented IIT Guwahati in **Improv comedy** among 23 IITs in Inter IIT Cultural Meet 6.0. and 7.0.
- Manthan:** Secured **Gold** medal in **Quite Sketchy** (Content creation) among 13 Hostels.
- Manthan:** Secured **Silver** medal in **Improv Comedy** among 13 Hostels.
- Core Team, IITG.Ai:** Taught **advanced AI concepts** and presented projects to industry experts for feedback.
- Coordinator, Coding Club:** Directed machine learning activities, fostering **collaboration** and **hands-on projects**.
- Member, Anchorenza and RadioG:** Organized fresher **introductory videos**, **Mr. & Ms. Fresher**, and **creative parody videos**.